MERN Backend Concepts Cheat Sheet – Ayush Verma

1■■ Middleware in Express:
• Middleware are functions that run between request and response.
• Used for authentication, logging, validation, and error handling.
• Example: fetchUser middleware verifies JWT token before route access.

2■■ How JWT Works Internally:
• JWT = Header.Payload.Signature
• Header defines algorithm, Payload contains user data, Signature verifies authenticity.
• Created using jwt.sign(), verified using jwt.verify().
• Stateless authentication – no session storage needed.

3■■ Role of ObjectId in MongoDB:
• Each MongoDB document has a unique ObjectId (_id).
• 12-byte structure (timestamp + random + counter).
• Used to identify or reference other documents (relations).

4■■ Relational Data in MongoDB:
• MongoDB is NoSQL but supports relationships using:
  - References (ObjectId + ref)
  - Embedded subdocuments
• Example: Each Note stores user's ObjectId to link Note ↔ User.

5■■ CRUD Operations (Quick Recap):
• Create – POST /addnote → note.save()
• Read – GET /fetchallnotes → Note.find()
• Update – PUT /updatenote/:id → findByIdAndUpdate()
• Delete – DELETE /deletenote/:id → findByIdAndDelete()

6■■ Async/Await Importance:
• Ensures database operations finish before sending response.
• Without it, code runs asynchronously and may send incomplete data.
• async/await replaces messy .then() chains and improves readability.

7■■ CORS & express.json():
• cors() → allows frontend (port 3000) to access backend (port 5000).
• express.json() → parses incoming JSON request bodies.

■ Tip: Use try/catch around all async routes to handle runtime errors safely.